

Euclid's Elements — Book XI

Proposition 24

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If a solid be contained by parallel planes, the opposite planes in it are equal and parallelogrammic.

1 Purpose and Scope

Proposition XI.24 begins a new phase of Book XI. The preceding propositions develop the geometry of perpendicular and parallel lines and planes, solid angles, and the realization of trihedral data. XI.24 is the first proposition in which this infrastructure is assembled into the geometry of a complete six-faced solid.

In modern language the configuration is a parallelepiped determined by three pairs of parallel planes. Euclid's conclusion has two parts:

1. every face of the solid is a parallelogram;
2. each pair of opposite faces is equal.

The formal reconstruction keeps these two tasks separate. The first is primarily an incidence theorem and is driven by XI.16. The second is metric: I.34 provides corresponding side equalities, XI.10 provides equality of the included spatial angles, and spatial SAS identifies corresponding diagonal triangles.

XI.24 is also particularly useful for comparing the direct Hilbert and Hilbert–Wyler routes. Both routes have the same input configuration, the same metric core, the same final result type, and the same audited kernel dependencies. They differ only in the incidence mechanism used to construct the section lines that form the six parallelogram faces.

2 Euclid's Statement

Euclid's statement is:

If a solid be contained by parallel planes, the opposite planes in it are equal and parallelogrammic.

The phrase “contained by parallel planes” is geometrically compressed. For XI.24 it means that the solid is bounded by three pairs of opposite parallel planes. The formal statement therefore makes this six-plane configuration explicit rather than leaving the combinatorics of the solid implicit.

3 Classical Configuration

Let

$$\pi_0 \parallel \pi_1, \quad \rho_0 \parallel \rho_1, \quad \sigma_0 \parallel \sigma_1$$

be the three pairs of parallel planes.

The eight vertices are indexed by choosing one plane from each pair:

	π	ρ	σ
A	π_0	ρ_0	σ_1
B	π_0	ρ_0	σ_0
C	π_0	ρ_1	σ_0
D	π_0	ρ_1	σ_1
E	π_1	ρ_0	σ_1
F	π_1	ρ_0	σ_0
G	π_1	ρ_1	σ_0
H	π_1	ρ_1	σ_1 .

Thus the six faces are

$$\begin{aligned} \pi_0 : ABCD, \quad \pi_1 : EFGH, \\ \rho_0 : ABFE, \quad \rho_1 : DCGH, \\ \sigma_0 : CBFG, \quad \sigma_1 : DAEH. \end{aligned}$$

The indexing is chosen so that every face is obtained by fixing one of the six planes and allowing the other two plane families to vary. This symmetry is exploited heavily in the Lean proof: only one face and one pair of opposite faces need a proposition-specific proof; the remaining cases follow by permutation of the three plane directions.

4 Euclid’s Classical Proof

4.1 Every face is a parallelogram

Consider the face $ABCD$ in the plane π_0 .

The parallel planes ρ_0 and ρ_1 are cut by the plane π_0 . Proposition XI.16 says that the two section lines are parallel. These section lines are AB and CD , hence

$$AB \parallel CD.$$

Likewise, cutting the parallel planes σ_0 and σ_1 by π_0 gives

$$BC \parallel DA.$$

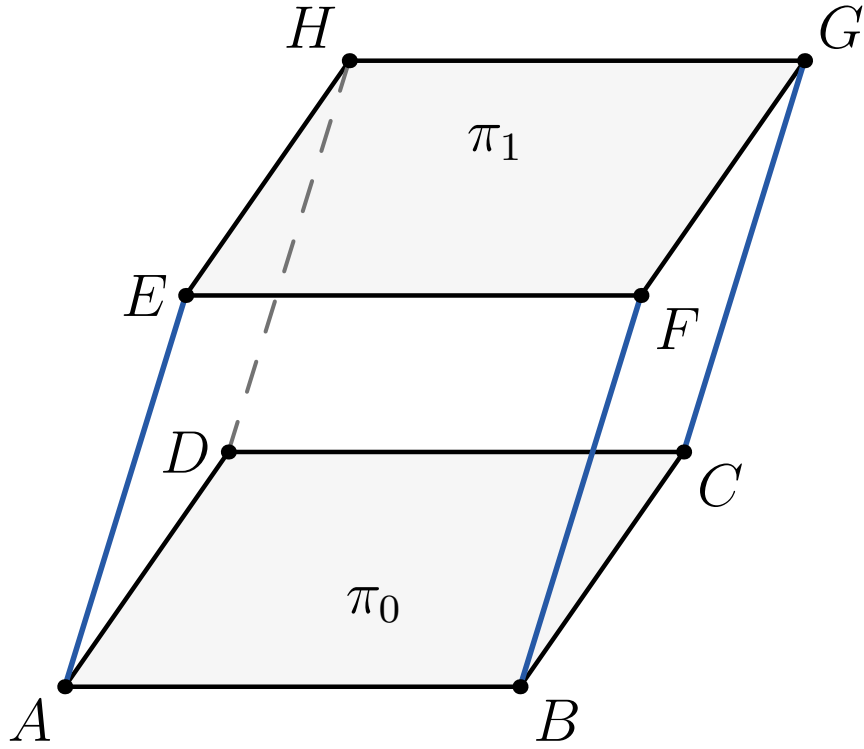


Figure 1: Parallelepiped configuration.

Therefore $ABCD$ is a parallelogram. The same argument applies to every other face after permuting the three pairs of parallel planes.

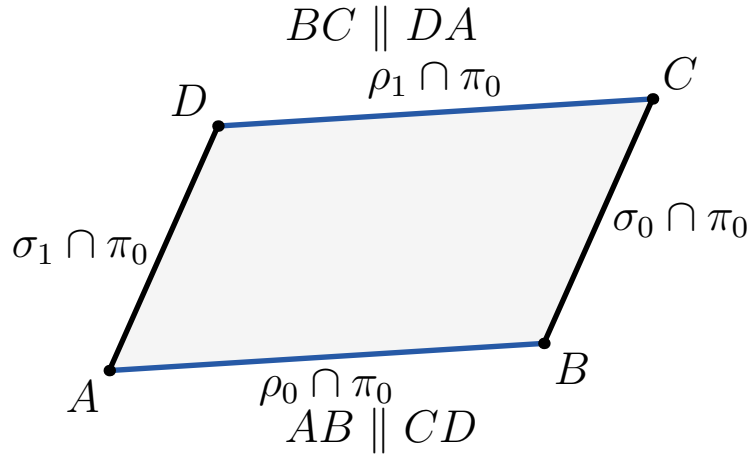


Figure 2: XI.16 on one face.

4.2 Opposite faces are equal

Consider the opposite faces $ABCD$ and $EFGH$. Euclid compares diagonal triangles in these faces.

The side face $ABFE$ is a parallelogram, so by I.34

$$AB \cong EF.$$

The side face $CBFG$ is also a parallelogram, so

$$CB \cong GF.$$

The lines forming the angle at B are parallel, in the appropriate directions, to the lines forming the angle at F . Proposition XI.10 therefore gives

$$\angle ABC \cong \angle EFG.$$

By I.4, the triangles

$$BAC \quad \text{and} \quad FEG$$

are congruent.

Euclid then passes from the diagonal triangles to the complete parallelograms: by I.34 each diagonal bisects a parallelogram into two equal triangles, so the two opposite parallelogram faces are equal. The same argument applies to the other two pairs of opposite faces.

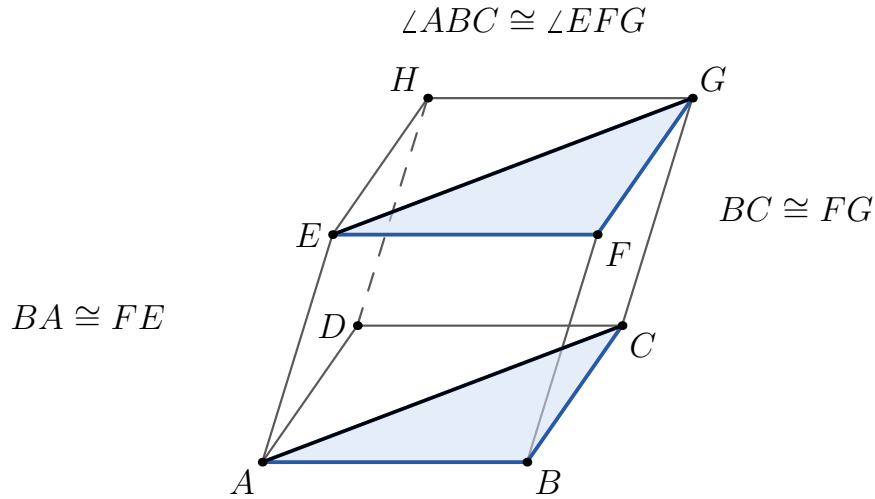


Figure 3: Opposite-face SAS.

5 Classical Dependency Pattern

The classical proof has a particularly transparent dependency structure.

First:

$$\boxed{\text{XI.16}} \longrightarrow \boxed{\text{six parallelogram faces}}.$$

Then, for one pair of opposite faces:

$$\begin{aligned} \text{I.34} &\implies \text{two corresponding side equalities,} \\ \text{XI.10} &\implies \text{included spatial angle equality.} \end{aligned}$$

Finally:

$$\boxed{\text{I.4 (SAS)}} \longrightarrow \boxed{\text{congruent diagonal triangles}} \longrightarrow \boxed{\text{equal opposite parallelograms}}.$$

No proportion theory is needed. No volume notion is needed. The theorem is a synthesis of parallel-plane incidence with ordinary congruence geometry.

6 What Is Genuinely Three-Dimensional Here?

The conclusion concerns six planar parallelograms, but XI.24 is not a planar theorem disguised as a solid theorem.

There are two genuinely spatial steps.

First, XI.16 converts parallelism of planes into parallelism of their section lines. The face parallelograms therefore arise from three-dimensional incidence, not from an independent planar assumption.

Second, XI.10 compares angles formed by pairs of corresponding parallel lines situated in different parts of the solid. This is the spatial bridge needed for the SAS comparison of opposite-face triangles.

By contrast, I.34 is used entirely within one face plane. The formal proof therefore treats I.34 through an induced **PlaneGeo** slice rather than installing planar Euclidean structure globally on the ambient three-dimensional geometry.

XI.24 is thus a useful example of the project’s basic architecture:

spatial incidence + local planar geometry + spatial congruence.

7 Formal Target and Statement

The input configuration is represented by

```
HilbertParallelepipedConfiguration
```

It stores:

- the three parallel-plane hypotheses;
- the twenty-four point–plane incidences locating the eight vertices.

The conclusion is represented by

```
HilbertXI24Conclusion
```

It contains six parallelogram fields:

```
face_pi0  
face_pi1  
face_rho0  
face_rho1  
face_sigma0  
face_sigma1
```

and three opposite-face triangle-congruence witnesses:

```
opposite_pi  
opposite_rho  
opposite_sigma
```

The direct public theorem is

```
Geometry.euclid_proposition_11_24
```

and the Hilbert–Wyler public theorem is

```
Geometry.euclid_proposition_11_24_wyler
```

Both have the same hypothesis type and the same conclusion type.

8 Spatial / Planar Architecture Used

The direct theorem lives in the established Book XI three-dimensional environment:

```
HilbertIncidence  
HilbertOrder  
HilbertSpacePrimitive  
HilbertSpaceIncidence  
HilbertSpaceOrder  
HilbertSpaceCongruence  
HilbertSpaceEuclidean
```

The proof deliberately avoids a global

```
HilbertEuclideanPlane Geo
```

instance on the ambient geometry.

Whenever a planar theorem is required, the relevant carrier plane is turned into an induced plane geometry. In particular, I.34 is used inside

```
PlaneGeo
```

and its congruence conclusions are transported back to the ambient space.

This separation is essential. It prevents a theorem about a three-dimensional configuration from silently inheriting planar Euclidean structure at the ambient level.

9 Proof Architecture of the Direct Reconstruction

9.1 Step 1: one face from XI.16

The reusable one-face theorem is

```
hilbert_XI24_face_parallelogram
```

For $ABCD \subset \pi_0$, it applies

```
euclid_proposition_11_16
```

twice.

The first application uses ρ_0, ρ_1 and obtains the two section lines carrying AB and CD . Their spatial parallelism is converted into the point-defined relation

```
Geo.Parallel A B C D
```

The second application uses σ_0, σ_1 and obtains

```
Geo.Parallel B C D A
```

Together these are exactly the two opposite-side parallelisms required for

```
IsParallelogram Geo A B C D
```

9.2 Step 2: deriving the endpoint inequalities

The point-defined relation `Geo.Parallel` contains nondegeneracy data. Thus the proof must establish

$$A \neq B, \quad C \neq D, \quad B \neq C, \quad D \neq A.$$

These inequalities are not added as hypotheses. Each follows directly from the disjointness of one of the relevant pairs of parallel planes. For example, if $A = B$, then the same point would lie on both σ_0 and σ_1 , contradicting $\sigma_0 \parallel \sigma_1$.

This is a typical example of hidden formal data that Euclid leaves implicit because the diagram already displays a nondegenerate solid.

9.3 Step 3: all six faces

The theorem

```
hilbert_XI24_all_faces_parallelograms
```

applies the one-face theorem six times.

The results are:

$$\begin{array}{ll} \text{Par}(A, B, C, D), & \text{Par}(E, F, G, H), \\ \text{Par}(A, B, F, E), & \text{Par}(D, C, G, H), \\ \text{Par}(C, B, F, G), & \text{Par}(D, A, E, H). \end{array}$$

No new geometry occurs here. The proof is a symmetry layer over one incidence argument.

9.4 Step 4: I.34 inside a face plane

The helper

```
hilbert_XI24_face_opposite_sides_congruent
```

takes an ambient parallelogram known to lie in a plane.

It performs three tasks:

1. realize the four vertices in the induced **PlaneGeo**;
2. invoke the planar I.34 theorem there;
3. transport the resulting segment congruences back to the ambient geometry.

For the first pair of opposite faces this gives, from the two side faces,

$$BA \cong FE, \quad BC \cong FG.$$

This is exactly the side data needed for SAS.

9.5 Step 5: directed parallelism for XI.10

XI.10 does not consume merely unordered parallel lines. It needs directed parallel-segment data.

The helper

```
hilbert_XI24_parallelogram_directed_opposite_sides
```

extracts the required direction from a parallelogram.

For the side faces $ABFE$ and $CBFG$, the proof obtains

$$AB \parallel_{\text{dir}} EF, \quad CB \parallel_{\text{dir}} GF.$$

This directed information is an important formal refinement of the classical phrase “the sides are parallel respectively.”

9.6 Step 6: the noncoplanarity hypothesis of XI.10

The six points used in the angle comparison must not all lie in one plane.

For the first opposite-face pair, suppose that

$$A, B, C, E, F, G$$

were contained in some plane ω .

Since A, B, C are three noncollinear vertices of the parallelogram $ABCD$, plane uniqueness forces

$$\omega = \pi_0.$$

Likewise the noncollinear triple E, F, G forces

$$\omega = \pi_1.$$

Hence $\pi_0 = \pi_1$, contradicting their parallelism.

Thus the noncoplanarity required by XI.10 is derived from the solid configuration rather than assumed independently.

9.7 Step 7: XI.10 gives the included angle

The directed side data and the noncoplanarity proof are passed to

```
euclid_proposition_11_10
```

The result is the ambient angle congruence

$$\angle ABC \cong \angle EFG.$$

This is the genuinely spatial metric step in the opposite-face comparison.

9.8 Step 8: spatial SAS

The theorem

```
hilbert_XI24_opposite_pi_faces_triangle_congruent
```

combines

$$BA \cong FE, \quad BC \cong FG, \quad \angle ABC \cong \angle EFG.$$

The ambient three-dimensional geometry does not use a global planar `HilbertCongruence` instance. Instead the proof invokes the spatial SAS consequences

```
hilbert_space_sas_third_side  
hilbert_space_sas_remaining_angles
```

The result is

```
TriangleCongruenceResult Geo B A C F E G
```

Thus the entire congruence of the two diagonal triangles is made explicit.

9.9 Step 9: the other two opposite-face pairs

Only the π_0/π_1 pair is proved metrically from scratch.

The theorem

```
hilbert_XI24_all_opposite_faces_triangle_congruent
```

constructs two relabeled parallelepiped configurations and reuses the same theorem for the ρ - and σ -directions.

The three witnesses are

opposite faces	triangle witness
π_0/π_1	$BAC \cong FEG,$
ρ_0/ρ_1	$BAF \cong CDG,$
σ_0/σ_1	$BCF \cong ADE.$

This permutation strategy is preferable to copying the metric proof three times: it exposes the actual symmetry of the solid.

9.10 Step 10: final assembly

The public theorem

```
euclid_proposition_11_24
```

has a deliberately short final body.

It obtains the six faces from

```
hilbert_XI24_all_faces_parallelograms
```

and the three opposite-face witnesses from

```
hilbert_XI24_all_opposite_faces_triangle_congruent
```

and then fills the fields of

```
HilbertXI24Conclusion.
```

This separation keeps proposition-specific construction out of the public wrapper.

10 Why the Formal Conclusion Uses Diagonal Triangles

Euclid’s final conclusion is equality of opposite parallelogram faces. Classically this is obtained by saying that each parallelogram is twice one of its diagonal triangles.

The present library does not introduce a general theory of area or equality of arbitrary plane figures merely to formalize this one sentence. Instead, the conclusion stores a stronger local synthetic witness: the corresponding diagonal triangles are congruent.

For the first pair this witness is

```
TriangleCongruenceResult Geo B A C F E G
```

which contains all three side congruences and all three angle congruences.

This choice has three advantages.

1. It stays inside the existing synthetic congruence language.
2. It avoids introducing numerical area or measure.
3. A future theory of equal plane figures or area can recover Euclid’s final doubling argument from the stored triangle congruence.

Thus the formal proof does not replace Euclid by coordinates or arithmetic. It stops one structural step before the classical area statement.

11 Hilbert–Wyler Reconstruction

XI.24 has a second production theorem:

```
Geometry.euclid_proposition_11_24_wyler
```

The important point is that this is not an alias of the direct theorem.

11.1 Wyler XI.16 as the incidence front end

The Wyler one-face theorem is

```
hilbert_XI24_face_parallelogram_wyler
```

Instead of the direct XI.16 theorem, it invokes

```
euclid_proposition_11_16_wyler
```

The Wyler XI.16 interface returns exact carrier identities for the section lines. Schematically,

$$\text{carrier}(\rho_0) \cap \text{carrier}(\pi) = \text{carrier}(l_{AB}),$$

and similarly for the opposite section.

The incidence $B \in l_{AB}$, for example, is recovered from

$$B \in \rho_0, \quad B \in \pi$$

by rewriting with the carrier equality.

Only after these exact meet identities are used does the proof convert the section-line result to the project's point-defined `Geo.Parallel` relation.

Thus the Wyler route changes the incidence language at precisely the place where XI.24 depends on XI.16.

11.2 Six faces through the Wyler route

The theorem

```
hilbert_XI24_all_faces_parallelograms_wyler
```

repeats the same carrier-based one-face construction under the six symmetric relabelings.

The result is the same six parallelogram faces as in the direct proof.

11.3 The metric core is intentionally shared

After the six faces have been established, the direct and Wyler proofs rejoin.

The Wyler route uses the same helpers for:

- I.34 in `PlaneGeo`;
- directed opposite sides of a parallelogram;
- XI.10;
- spatial SAS.

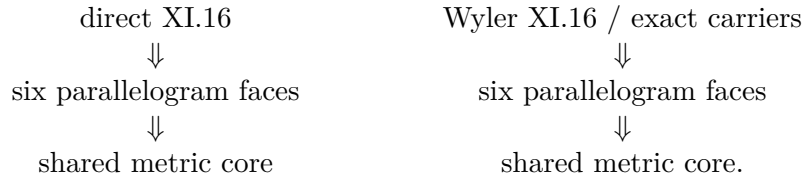
Its one-pair theorem is

```
hilbert_XI24_opposite_pi_faces_triangle_congruent_wyler
```

and the all-pairs theorem is

```
hilbert_XI24_all_opposite_faces_triangle_congruent_wyler
```

The two proof spines are therefore:



This is exactly the intended role of the Hilbert–Wyler program: alter the incidence interface where it matters without duplicating a successful metric library.

12 Lean Formalization

The production direct file is

```
CGJteamLab/Proposition11_24.lean
```

Its principal declarations are:

```
hilbert_XI24_face_parallelogram
HilbertParallelepipedConfiguration
hilbert_XI24_all_faces_parallelograms
hilbert_XI24_parallelogram_directed_opposite_sides
hilbert_XI24_face_opposite_sides_congruent
hilbert_XI24_opposite_pi_faces_triangle_congruent
hilbert_XI24_all_opposite_faces_triangle_congruent
HilbertXI24Conclusion
euclid_proposition_11_24
```

The production Wyler file is

```
CGJteamLab/Wyler/Proposition11_24.lean
```

Its principal declarations are:

```
hilbert_XI24_face_parallelogram_wyler
hilbert_XI24_all_faces_parallelograms_wyler
hilbert_XI24_opposite_pi_faces_triangle_congruent_wyler
hilbert_XI24_all_opposite_faces_triangle_congruent_wyler
euclid_proposition_11_24_wyler
```

The final full project build is clean.

13 Hidden Formal Data

XI.24 is visually simple but formally dense. Several pieces of information that are immediate from Euclid’s diagram must be made explicit.

13.1 The solid is not determined by the phrase “parallel planes”

The formal theorem needs six named planes, grouped into three parallel pairs, and eight named vertices with explicit incidence data. Without this combinatorial information there is no formal notion of which four vertices form each face.

13.2 Face edges must be nondegenerate

The relation `Geo.Parallel A B C D` requires $A \neq B$ and $C \neq D$. These facts are derived from disjoint parallel planes rather than assumed separately.

13.3 Section-line membership must be reconstructed

XI.16 returns line-level section data. XI.24 must prove that the other vertices of a face lie on those same section lines. In the direct route this uses the section characterization supplied by XI.16. In the Wyler route it uses exact carrier equalities.

13.4 Planar and ambient congruence are different layers

I.34 is a planar theorem. XI.24 must enter `PlaneGeo`, apply I.34, and transport the resulting segment congruences back to the ambient three-space.

13.5 Parallelism for XI.10 has direction

The classical wording “parallel respectively” suppresses orientation. The formal XI.10 interface requires directed parallel segments, so the orientation of the opposite sides of each parallelogram must be recovered explicitly.

13.6 Noncoplanarity is a theorem, not a picture

The two opposite face triangles cannot simply be declared to occupy different planes. Their noncoplanarity is proved using noncollinearity of parallelogram vertices, plane uniqueness, and the parallelism of the two opposite face planes.

13.7 Equality of faces has no primitive area meaning

The current library has no general primitive relation saying that two parallelograms have equal area. XI.24 therefore stores congruent diagonal triangles as the synthetic equality witness.

14 Relationship with XI.10 and XI.16

XI.24 is the first proposition that combines XI.16 and XI.10 in a single solid-geometry theorem. XI.16 provides the face structure:

$$\text{parallel planes} \implies \text{parallel section lines} \implies \text{parallelogram faces.}$$

XI.10 provides the metric comparison between different parts of the solid:

corresponding directed parallel lines $\implies$ equal included spatial angles.

Thus the two earlier propositions have complementary roles. XI.16 is the incidence engine; XI.10 is the angle-transport engine.

I.34 and I.4 then supply the planar and metric closure needed to identify opposite faces.

15 Relationship with the Book XI Architecture

XI.24 is an integration theorem.

It consumes established infrastructure rather than adding a new primitive geometric notion:

- parallel-plane incidence from XI.16;
- spatial angle transfer from XI.10;
- local `PlaneGeo` geometry;
- ambient segment and angle congruence;
- plane uniqueness and noncollinearity infrastructure.

It also introduces a reusable explicit model of a parallelepipedal configuration:

```
HilbertParallelepipedConfiguration
```

which can serve as a stable interface for later propositions involving parallelepipeds and parallelepipedal solids.

In this sense XI.24 marks a transition from local line–plane results to structured solid configurations.

16 Relationship with the Foundations

No coordinate model is used.

No vector space is used.

No numerical length function is used.

No numerical angle measure is used.

No area or volume function is used.

The direct proof relies on the project’s established Hilbert incidence, order, congruence, and Euclidean layers. Planar Euclidean reasoning is localized to explicit `PlaneGeo` slices.

The Wyler proof does not introduce a second metric foundation. It changes only the incidence route used to obtain face section lines. The downstream metric argument remains the same synthetic Hilbert argument.

17 E^n Transfer Audit

XI.24 is not yet formalized as a theorem of arbitrary-dimensional Euclidean geometry.

Nevertheless, its structure separates the dimension-sensitive and dimension-local pieces quite clearly.

The face construction requires an incidence theorem saying that suitable parallel flats cut by another flat produce parallel section lines. The Wyler XI.16 route already expresses this step through carrier intersections, which is closer to the language needed for dimension-independent transfer.

The metric part of the proof is more strongly three-dimensional in the current library:

- XI.10 is presently formulated in the three-dimensional Hilbert environment;
- the noncoplanarity argument uses uniqueness of the two opposite face planes inside that environment;
- spatial SAS is provided by the established 3D congruence layer.

Accordingly, XI.24 currently proves:

Hilbert 3D $\implies$ direct XI.24 and Wyler-routed XI.24.

It does not yet prove:

$E^n \implies$ XI.24 inside an arbitrary ambient dimension.

A future E^n transfer would naturally seek a local three-dimensional flat containing the parallelepipedal configuration and then reuse the present metric proof inside that flat.

18 Axiomatic Status

The kernel audit of the direct theorem

```
Geometry.euclid_proposition_11_24
```

reports:

```
[propext,  
 Classical.choice,  
 Geometry.bookZero_nullSegment1,  
 Quot.sound]
```

The kernel audit of the Wyler theorem

```
Geometry.euclid_proposition_11_24_wyler
```

reports exactly the same set:

```
[propept,
  Classical.choice,
  Geometry.bookZero_nullSegment1,
  Quot.sound]
```

Thus:

New proposition-specific axiom:	no
New spatial axiom:	no
bookZero_nullSegment1:	yes
bookZero_nullSegment2:	no
Circle-circle continuity:	no
Numerical angle measure:	no
Coordinates or vectors:	no
Global ambient planar Euclidean instance:	no
Explicit PlaneGeo slices:	yes
sorry / admit:	no
Full lake build:	clean

The equality of the two audit profiles is significant. The direct and Wyler proofs are genuinely different at the incidence layer, but the change in proof organization has no additional kernel cost.

XI.24 also differs sharply from XI.23: it does not depend on

```
Geometry.bookZero_nullSegment2
Geometry.hilbert_circle_circle_intersection
```

and hence does not cross the circle-circle continuity boundary.

19 Diagrammatic Audit

Three stable geometric states are retained.

1. The complete parallelepipedal configuration with the eight vertices and the two displayed opposite face planes.
2. A single face together with the four section lines obtained from the two relevant parallel-plane pairs.
3. The two opposite diagonal triangles used in the SAS comparison.

No separate figure is introduced for:

- each of the remaining five face-parallelogram constructions;
- each application of I.34;

- the directed-parallel extraction;
- the plane-uniqueness contradiction used for noncoplanarity;
- the two relabelings used for the remaining opposite-face pairs.

Those operations change logical information but do not create a new stable geometric state.

The figure convention follows the established Book XI style: planar geometry is black or gray; genuinely spatial connectors may be emphasized; hidden spatial continuation may be dashed; plane or face fills are light.

20 Dependency Summary

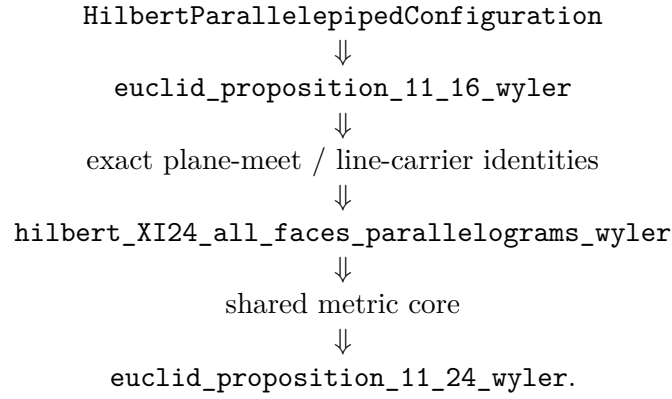
The classical local chain is:

three pairs of parallel planes
 $\Downarrow$
 XI.16
 $\Downarrow$
 six parallelogram faces
 $\Downarrow$
 I.34 + XI.10
 $\Downarrow$
 I.4 (SAS)
 $\Downarrow$
 congruent diagonal triangles
 $\Downarrow$
 equal opposite parallelograms.

The production direct Lean chain is:

HilbertParallelepipedConfiguration
 $\Downarrow$
 hilbert_XI24_face_parallelogram
 $\Downarrow$
 hilbert_XI24_all_faces_parallelograms
 $\Downarrow$
 PlaneGeo / I.34 + directed parallel segments + XI.10
 $\Downarrow$
 spatial SAS
 $\Downarrow$
 hilbert_XI24_all_opposite_faces_triangle_congruent
 $\Downarrow$
 euclid_proposition_11_24.

The production Wyler chain differs only at the incidence front end:



21 Conclusion

Proposition XI.24 is the point at which the theory of parallel planes becomes the geometry of a structured solid.

The formal reconstruction preserves Euclid’s global proof architecture very closely. XI.16 produces the six parallelogram faces; I.34 provides corresponding side equalities; XI.10 transfers the included spatial angle; and SAS identifies the diagonal triangles of opposite faces.

The main formal refinements concern data that the classical diagram suppresses: the six-plane combinatorics of the solid, endpoint distinctness, exact section-line membership, the distinction between planar and ambient congruence, direction on parallel sides, and the proof that opposite face triangles are not coplanar.

The Hilbert–Wyler version makes the architectural separation especially clear. Its incidence front end is genuinely different: XI.16 is routed through exact carrier intersections. Once the six parallelogram faces are available, however, the proof rejoins the direct route and uses exactly the same metric core.

The final audit therefore gives a particularly clean result:

different incidence organization + shared synthetic metric core $\implies$ same theorem and same axiom pro

XI.24 is consequently not only a reconstruction of one Euclidean proposition. It is a compact integration test for the mature Book XI architecture and one of the clearest production examples of the project’s Hilbert–Wyler strategy.